

Object Tracking using OpenCV with Python

Business Objective

Object tracking is the task of automatically identifying objects in a video and interpreting them as a set of trajectories with high accuracy. Object tracking is an application of deep learning. It takes an initial set of object detections, creates a unique ID for each of the initial detections, and then tracks each object as they move around frames in a video, maintaining the ID assignment.

Object tracking can be done on two levels, Single Object Tracking (SOT) and Multiple Object Tracking (MOT). In single object tracking, only a single object is tracked even if the environment has multiple objects in it. The object to be tracked is determined by the initialization in the first frame. Whereas in multiple object tracking, all the objects present in the environment are tracked over time. If a detection-based tracker is used, it can also track new objects that emerge in the middle of the video.

Data Description

In this particular project, we will use various video clips to perform object tracking.

Aim

To perform single object tracking (SOT) and multiple object tracking (MOT) on a given input video using various algorithms.

Tech stack

- Language - Python
- Libraries -cv2, numpy, os, moviepy

Approach

1. Create a new virtual environment in your terminal
2. Importing the required libraries.
3. Single Object Tracking
 - Define a class for single object tracking
 - List the required algorithms that need to be implemented
 - Run the single object tracker object class
 - Select the desired object in the video and press enter
4. Multiple Object Tracking

- Create a Multiple object tracking class
 - List the required algorithms that need to be implemented
 - Run the multiple object tracker object class
 - Select the desired objects in the video and press enter
5. Centroid tracking algorithm
- Create a centroid algorithm class
 - Read the input video for face detection
 - Download the Caffe model
 - Execute the model to detect the faces from the input video
6. Stacking results
- Import the required dependencies
 - Create a single object tracking class
 - List all the required algorithms
 - Create a function for stacking videos
 - Create a stacked video using the MoviePy module

Modular code overview

InputFiles

```
|_car_driving.mp4
|_chaplin.mp4
|_horse_riding.mp4
|_run.mp4
```

SourceFolder

```
|_Engine.py
|_ML_Pipeline
    |_CentroidTracking.py
    |_SingleObjectTracking.py
    |_MultiObjectTracking.py
    |_deploy.prototxt
    |_res10_300x300_ssd_iter_140000.caffemodel
    |_goturn.caffemodel
    |_goturn.prototxt
|_README
|_requirements.txt
```

Lib

```
|_StackingTrackingResults.ipynb
|_Reference
    |_ObjectTracking ppt
```

OutputFolder

```
|_video.mp4
```

Once you unzip the modular_code.zip file, you can find the following folders within it.

1. InputFiles
2. SourceFolder
3. OutputFolder
4. Lib

1. Input folder - It contains all the data that we have for analysis. We have 4 input videos present.
2. Source folder - This is the most important folder of the project. This folder contains all the modularized code for all the above steps in a modularized manner. This folder consists of:
 - Engine.py
 - ML_PipelineThe ML_Pipeline is a folder that contains all the functions put into different python files, which are appropriately named. These python functions are then called inside the engine.py file.
3. Output folder – The output folder contains the video generated after you run the engine.py file.
4. Lib folder - It contains the original ipython notebooks that we saw in the videos. There is a reference folder that consists of the presentations used during the explanation.

Project Takeaways

1. What is object tracking?
2. Differentiation between object tracking and object detection
3. Object tracking applications
4. Boosting Tracker algorithm
5. Multiple Instance Learning Tracker (MIL) algorithm
6. Kernelized Correlation Filters Tracker (KCF) algorithm
7. Tracking, Learning, Detection Tracker (TLD) algorithm
8. Median Flow Tracker algorithm
9. Generic Object Tracking Using Regression Networks Tracker (GOTURN) algorithm
10. Minimum Output Sum of Squared Error Tracker (MOSSE) algorithm
11. Discriminative Correlation Filter Tracker (CSRT) algorithm
12. Centroid Tracking algorithm
13. How to implement Single Object Tracker (SOT)
14. How to implement Multiple Object Tracker (MOT)
15. What are the challenges in object tracking?
16. How to stack multiple algorithm results in a single video.